Corrections and Clarifications

FFGFT / T0 Time-Mass Duality

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Contents

Preliminary remark

This document is the compact, running correction register of the FFGFT corpus. It supersedes the previous extended edition, which remains available unchanged as **Doc. 190-Archive** (`190_T0_Korrekturen_Archiv_De/En`) and contains, for every entry, the full justification, the faulty and corrected expressions verbatim, and all addenda. The table below lists *all* entries (C/K, P, R) with the affected documents and the subject in short form. The register remains append-only: source documents are not revised (cf. R50); new entries are appended here, and their elaboration will henceforth be written directly in this document.

.1 Register table

C/K = correction, P = clarification, R = revision/register entry. Full texts: Doc. 190-Archive.

No.	Affects	Subject (short form; details: Archive)
C1	049 (HTML)	$\xi = \lambda_h^2 v^2/(16\pi^3 m_h^2)$ *[Addendum 28 July 2026, cf. R66: reason for the correction — $16\pi^3 = (4\pi)\cdot(4\pi^2) = S^2 \cdot 2S^3$ is the sphere count of the compact T^4 boundary geometry (Doc. 310); the earlier version counted one 2-sphere too many]*
C2	116 (tab.)	$r_\tau = 25/9$
C3	018 (Explorer)	$a_e = 2\pi^2/(f \cdot k)$
C4	267 (DE+EN), 268 (DE)	**Factor 3 = three observable space dimensions:** the observer measures $k^2_{\text{obs}} = (n_1^2 + n_2^2 + n_3^2)/L^2$ (the fourth torus direction is time unfolding).
C5	275 (script v3)	**All three fixed in v4** (11 June 2026): T^4 as kNN in genuine 4D coordinates with dimension-matched 4D nulls (result: $1{,}329 \pm 0{,}025$ *inside* the band $[0{,}690,\ 2{,}406]$); HLV ensemble with $\sigma = 0{,}05$ ($0{,}710 \pm 0{,}008$, in the band); the shell detect ...
P1	116 / 158	both correct; $0{,}001\,\%$ for external use
P2	several	$\hbar$ follows from ξ via $L_0 = \xi \cdot \ell_P$
P3	173–176	two parameters: ξ_0 (geom.) and ξ_{Higgs} (alg.)
P4	180–182	L_0 = geom. fundamental length
P5	several	mass structure from ξ; $\sim 1\,\%$ agreement
P6	116 / 158	Koide only with numerical values (non-closure, Doc. 193)
P7	005, 008, 019, 086, 189, 202	scale structure from the recursion $\mathscr{R}$
P8	005, 008, 012, 014, 041, 060, 133, 141, 159, 181	geometric scale adjustment / fractal correction
P9	005, 006, 042, 046	columns “scaling class” + “SM correspondence”; quarks via $m = r\xi^p v$
P10	009	$K = R_{pe}/(4/3)^7$: an empirical anchor; factor $1{,}314$ defined implicitly, not derived
P11	025, 003, 133	correct: $(1 - 275/4 \cdot \xi)$, $\Delta = 0{,}0002\,\%$; Doc. 003 and Doc. 133 use different models
P12	022, 230	two different observables: cumulative (Doc. 022) vs. elementary $\xi/(2\pi)$ (Doc. 230); 10^{-5} unreachable at any finite N from Doc. 022
P13	013, 025, 061	$\xi_{\text{FFGFT}}/\xi_{\text{UIFT}} = m_e c^2/(\frac{16}{9} \ln 2 \cdot \text{eV}) \approx 414\,684$; table in Doc. 013 (archived) reconstructed

No.	Affects	Subject (short form; details: Archive)
P14a	041	fractal path lengthening (geometric path stretching); energy loss only an artifact when the path lengthening is omitted; path-length dependence remains (Doc. 026/149/182); the same path lengthening carries the CMB treatment in Doc. 268 (formerly tracked separately as P19)
P15	061 / 041	marked as a ΛCDM comparison quantity; the static universe has no cosmological z; CMB by stationary thermalization
P16	026 / 064 / 181	H_0 is emergent (decompactified), not fundamental; implemented in Doc. 263, Docs. 026/064/181 noted as outdated
P17	028	the DE/DM relation is circular (gives 2.5 for any ξ); MOND partially derived ($\xi^{1/4}$ genuine, K_M free); CMB relations unaffected; noted
P18	267	the two pictures are *not* distinguishable through the observed ratios; the decision is theoretical, not empirical (conceptual placement)
P20	267 / 268	the *form* is fixed: a dimensionless ξ ratio (R_H/ℓ_P).
P29	268	*Only partially answered.* The factor 3 from the 3D projection ($3 \cdot \{1, 6, 14, 26\}$) is plausible, but the sequence $\{1, 6, 14, 26\}$ is *read off the data* (Doc. 268, step 4), not produced forward.
P30	268	the *naive* spherical Bessel sum $C_\ell = \sum_n N_{\mathrm{BE}} \lvert j_\ell(k_{\mathrm{obs}} D_A)\rvert^2$ does *not* reproduce $\{3, 18, 42, 78\}$ – it washes out to a quasi-continuum dominated by low modes.
P31	268	*Leading behaviour derived forward.* Forward, T^4 gives the harmonic backbone $1 : 2 : 3 : 4 : 5$ (symmetric resonance $k^2 = 3n^2$).
P32	corpus-wide	follows from P20: H_0 is an *imported* external calibration, not derived.
P33	009 / 042 / 181	the *form* of ξ is forward: the harmonic $4/3$ (perfect fourth, Doc. 042/159) gives the leading digit, 3 (space dimensions) and 2^2=4 (spacetime) are read off the geometry.
P34	181 / 013	*3+1 is empirical input* – no theory derives the dimension count; the earlier "follows necessarily" claim (Doc. 181) is overstated.
P35	corpus-wide (esp. 182, 250s, scale tables)	ξ^N **triviality:** "$X = \xi^N$" is by itself *vacuous*, since any magnitude can be written as a ξ power – an exponent alone is not a prediction (cf. P10, P20, P33: form forward, number input).
P36	182 / cosmol. sector	R_H **interpretation refined:** $R_H = c/H_0$ is the *Hubble length* – the local expansion rate expressed as a distance – *not* the size of the universe and *not* the observable horizon (the latter $\approx 3\,R_H$ in ΛCDM).
P37	009, 188, 271, 272, 275	**Three levels clearly separated:** **(1) Microscopic – input:** $\xi = 4/30000$ (fractal dimension).
P38	275	**Status clarification:** (a) $1/\varphi$ is a *pass-through point*, not a fixed point – the only fixed point of $r \mapsto r(1-\xi)$ is 0.
P39	182, 267, 277, corpus-wide (H_0/R_H)	H_0**/**R_H **status refined (shared reference point, not an FFGFT gap):** the relation Planck length↔ R_H (exponent $41/4$, $R_H/L_0 = 6{,}49 \times 10^{64}$) is derived from first principles by *no* framework – why the universe is $\sim 10^{60}$ Plan ...
C6	004, 025, 030, 039, 041, 053, 061, 063, 068, 081	**Redshift is achromatic.** The fractal path-lengthening is purely geometric and stretches *all* wavelengths by the same factor: $1 + z = e^{\xi x}$, independent of λ.
P40	005, 006, 011, 012, 133, 275	**Only ratios are exact.** Absolute values carry the fractal/SI correction, absorbed into the bridge constants ν (masses $m = r\xi^p\nu$), $E_0 = 7{,}398$ MeV (α) and $3{,}521 \times 10^{-2}/2{,}843 \times 10^{-5}$ (G; `calc_De.py`); in ratios they cancel ($m_\mu/m_e = 206{,}768$ ...

No.	Affects	Subject (short form; details: Archive)
P41	267, 280, 221 (cosmological sector)	**Checked and not booked (degeneracy risk at the shared scale).** ΛCDM signal $\Delta v = c\,\dot{z}\,\Delta t/(1+z)$ over 25 years: $+6$ cm/s (z=0.5), sign change at $z_0 = 1.92$, -14 cm/s (z=4), -20 cm/s (z=5); ELT/ANDES order $\sigma \sim 2$ cm/s.
P42	282, 292 (lepton sector)	**One declared reference point: the ratio m_μ/m_e.** The framework fundamentally keeps the purely theoretical derivation from ξ; the lepton sector receives – analogous to P39 (H_0/R_H as shared empirical reference) – exactly *one* declared reference point …
P43	006, 011, 292 (`calc_De.py`)	**No rework; recorded as a note.** Verified: `calc_De.py` does *not* use $K_{\mathsf{frak}} = 1 - 100\xi$ explicitly.
R41	284 (cf. 270, 249)	**The loss is not the SI conversion.** Per Doc. 270 the Type I duality decompactification $S^1_m \to \mathbb{R}_t$ ($\tilde{T}\cdot m = 1$) is a *mode-preserving embedding* (lossless); compactness survives as the discrete spectrum and as physical observables (T_{rev} …
R42	270 (cf. 248)	**Not an open geometric question – it is the measurement.** In the compact view the four circles are symmetric (no space/-time distinction); time is not selected by the geometry but is *relational* – the symmetry-breaking *is* the measurement (the Type I decompact …
R43	285, 283	**Corrected to containment.** $C_3 < A_5$ (3-fold and 5-fold both permute the same icosahedron, `recompute_reconcile.py`) and $T^7 = T^4 \times T^3$ ($7 = 4+3$), so HLV ⊃ FFGFT at the group/dimension level – the models are reconcilable, not inco …
R44	272, 190 (P35), 285	**Sharpened (no contradiction).** φ acquires meaning *not* inside FFGFT but as the derived eigenvalue of HLV's 5-fold / perp sector (golden inflation $\varphi^2 = \varphi + 1$) – a marker of the boundary HLV∖FFGFT.
R45	285, 282–284	**Sharpened.** A quasicrystal has two spectra: the *structure / diffraction* spectrum is pure point (sharp Bragg peaks, $\approx 21\times$ above a random null, `quasicrystal_information.py`) – that is the information and exactly HLV's substrate-distinguishabilit …
R46	296 (cf. 025, 026, 063, 156)	**Open check-point (not yet processed).** This intermediate-scale formulation potentially runs *counter* to the already established corpus line, which treats dark matter/energy not as an open signature but as a *derived consequence*: Doc. 025 reads dark matter as a …
R47	285, 297, 298 (cf. 286, 284, 295)	**In the time sector: containment, not enlargement.** In unified notation – both $D6+$time, Marcel as $D6+$spiral-time, FFGFT in Hilbert space as $D6+$recursion-time – the spiral-time window is not additive but *internal*: a bounded section (window) of the unrolled re …
R48	304 (cf. 295, 301, 303)	**Note withdrawn, document made precise (not repaired).** The overstatement lay solely in the *mail* and was withdrawn without reservation; the document was written throughout from the FFGFT reading and already booked the bridge as restricted.
R49	304, 305 (cf. 295, 303, 272)	**Vocabulary cited, marked as analysis, abstract desymbolised — notation not renamed.** A consolidated first-use footnote attributes U1/U2/U3 *and* M_τ/M_B to Krüger (2026) with the primary source (`doi:10.5281/zenodo.21302278`) and documented lineage (Me …
R50	025, 063, 061, 064, 078 (superseded by 306; cf. 003, 010, 295, 298, 302)	**Justification replaced, conclusion unchanged.** The conclusion (no singular beginning; a finite core with a minimal scale L_0 set by ξ) stands.

No.	Affects	Subject (short form; details: Archive)
R51	049 (cf. 095, 306, 267, 302; notation)	**A notational relic, not a substantive contradiction — and nonetheless no longer authoritative.** *(a) The resolution:* in the original T0 view, T_0 never denoted the instant zero but the *smallest possible time interval*.
R52	101, 165 (numerical values; cf. 180, 182, 013, 250, 290)	**Values corrected; the main relation is unaffected.** *(i) Wrong* L_0: correct is $L_0 = \xi \ell_P = \frac{4}{30000} \cdot 1.616 \times 10^{-35}$ m = $\boxed{2.155 \times 10^{-39}\text{ m}}$ — as in Doc. 180 (the dedicated derivation), Docs. 013, 181, 182, 187 ...
R53	267 §518 (table row *source of the scale*; cf. 182, 165, P20/P35/P36)	**Corrected reading — and the symmetry becomes sharper, not weaker.** *(a)* On the FFGFT side, the table row *source of the scale* correctly reads not "from ξ" but **"set / calibrated to H_0"**.
R54	292, 306 (cf. 307)	**(a) N_0 is not a base unit.** $N_0 \approx 38.6$ is a *ladder-internal*, generation-counting factor ($N_g = g N_0$, Dok. 292); its value is *open* (P35: $100/e$, $100/\varphi^2$, 39 do not hit it reliably).
R55	292 §Cross-ladder coupling (cf. R54, P35, P40/P41)	**The law stands; its evidential breadth is one entry, not three.** *(i) The third entry is forced arithmetically.* From $m_\tau/m_e = (m_\tau/m_\mu)(m_\mu/m_e)$ a uniformly multiplicative correction satisfies $K^{p_3} = K^{p_1}K^{p_2}$, hence $p_3 = p_1 + p_2$ *identicall ...*
R56	188 §Special role of $n = 3$ (cf. 189 §Derivation hierarchy, 248, 230/231/232, R55)	**The three elements are hereby booked uniformly as foundational assumptions.** *(i) The declared framework.* FFGFT posits three structural elements: the fundamental relation $\tilde{T} \cdot m = 1$ (Doc. 189, level 1), the topology T^4 (Doc. 248), and the order Z_3 ...
R57	A015 (cf. 020, 030, A-Series)	**Demand dissolved — ξ is a fundamental parameter.** A proof is neither possible nor necessary; the demand was wrongly posed.
R58	A142, A140 (cf. 127, 163)	**Gravitational quantisation not required.** $G = \xi^2/(4m_{\text{char}})$ is intrinsic to the torus geometry; a separate quantum gravity does not arise, because the geometry T^4/Z_3 is pre-ordered and not dynamical in the same sense as quantum-field-theoretic fields.
R59	A130 (cf. 011, 043, 101; P40)	**Structurally explained [S].** Two causes: (i) SI values are scheme-dependent (1-loop, $\overline{\text{MS}}$), i.e. the EFT precision values are not theory-independent raw data; (ii) fractal corrections in v are not cleanly separable from the loop structure of the SM ...
R60	A160, A165 (cf. 020, 022, 023, 230; P35)	**Both factors now geometrically derived [B].** (i) 2π is the circumference of the unit circle — the natural conversion between the torus winding phase and the measurement angle; $\xi/(2\pi)$ is thus the phase correction per measurement.
R61	174, 175 (cf. A142, 147, 162; R57, P35)	**Data-driven value selection — not a second parameter; withdrawn.** *(i) The genesis:* ξ_{Higgs} was selected *because* it lands in the measured window; the two-spaces doctrine is rationalisation after the selection, not independent justification.
R62	005 §Baryon worked example (proton) (cf. R50, P35)	**The anchor of the formula line is $\pi^2/2$, not m_μ.** *(i) Diagnosis.* The missing factor identifies the error as a single symbol: $(\pi^2/2)/m_\mu = 46.705$ hits the missing factor to $0.07\,\%$.

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R63	253 §ξ resonance heuristic (code: `rsa/factorization_benchmark_library.py`; cf. 024, 075, 076, 176; R57, P35)	**ξ is inert in the "optimal" range; the optimisation switched the filter off.** *(i) Inertness.* The code threshold is $1/1000$.
P44	253, 190 (R63); cf. A135, P35	**Pre-registered, tested, and refuted for this design; the window width is a property of the measurement setup.** *(i) Design.* Shor phase estimation, 12 counting qubits (phase resolution $2^{-12} = 2.44 \times 10^{-4}$), 8 shots, $10\,\%$ fully random outcomes, 300 …
R64	`2/python/t0_no_fallback_demo.py`	**The procedure is right, the separating cut too coarse, and the label wrong.** *(i) What holds.* The counters `t0_success_count` and `fallback_success_count` are incremented at separate places (l. 376 resp. 280/300), `get_statistics()` report …
R65	024, 075, 076, 147, 176, 253 as well as the	**No procedure distinguishable from Shor exists; the method claim is withdrawn.** *(i) What is withdrawn.* (a) The claim of an independent T0 resp. ξ factorisation procedure.
R66	097 (DE/EN and the standalone versions under	**The $16\pi^3$ count is right — but not for the reason given in Doc. 097.**
C7	the fourteen HTML pages under `rsa/`	**Rebuilt, not annotated: the pages now say what they do.** No correction box was inserted; what was corrected is the computation, the labelling and the method text.
R67	A040 (fractal correction), A270 (bulk exponent 36),	*(i)* With $D_f^{\text{eff}} = 2.973$ (three decimals) the rounding interval leaves $n \in [98.3,\ 102.0]$ open; even $D = 2.973$ exactly gives $n = 100.12$.
R68	279 (H_0 chain), 308 (Λ reading,	*(i) Identification:* the Λ function is carried by the derived quantity $\Lambda^* := 1/R_H^2 = (\pi/2)^2\xi^{20}/\bar{\lambda}_e^2 = 5.218{\times}10^{-53}$ m^{-2} ("closure scale").
R69	308 §a_0 derivation; cf. P20, R68;	*(i)* From $\partial\mathrm{T}^4 = \varnothing$ (Doc. 306) an isotropic torus yields a compact time cycle $\tau_c = 2\pi/H_0 = 91.9$ Gyr [K|P20]; the energy quantum is exactly $\hbar H_0$, identical to the momentum quantum of the spatial cycles.
R70	061 (temperature/CMB), A085 (natural units),	T_0 scale gap
R71	309 (scale anchor, inheritance problem);	sharpening: side-taking in the Hubble tension
R72	A010 (glossary), A080 (units), A265	E_0: bare and corrected
R73	026 (geometric cosmology), 279	H_0 notations, exponent $41/4$
R74	025 (Cosmology), 063 (Cosmic), DE+EN each,	lithium problem: mechanism, not solution
R75	024, 034, 075, 076, 147, 176: ξ name collision and rewrite of the Shor documents	In all affected documents the search parameter is now denoted σ and placed as a sampling grid without physical meaning.
R76	318	Scope of mass derivations
R77	041, 160, 318	Classification of early coupling constant formulas

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R78	160, 005	α_s and running coupling
R79	321, 319, 049	Algebraic derivation of the $SU(3)_c$ gauge group
R81	323, 321, 160, 320	Weinberg angle at M_Z from ξ
R82	322, 320	Hilbert-space embedding and spectral theory
R83	319, 321, 323, 322, 320	Status update: closed bridges
R84	324, 321, 009	Casimir derivation of ξ; ξ not a G(6) spectral value
R85	313, 325, 257, 321, 322	Hawking mechanism closed microscopically
R86	Docs. 180, 322, 314, 321, 325	Self-adjointness of $\hat{F}$ fully closed; $\xi = \lambda_{\min}(\hat{F}_{D_4})$ **[B]**
R87	Docs. 320, 322, 340	Δm^2_{32}: numerical values corrected, dev. $+16.0\,\%$; candidate analysis added **[K]**. *Closed by Doc. 340 (R103): Galois derivation $\Delta m^2_{\text{atm}} = 120\, m_\nu^2$ from $GF(27)^*$ orbits, dev. $1.0\,\%$ **[K]**.*
R88	Docs. 326, 329	n_{thresh} scale-dependent: $n(L) = L/(\sqrt{2}\ell_P)$; Matzke's scale $= 2\pi\ell_P/\ln 2$ **[B]**, universal number **[X]**
R89	Docs. 180, 019, 201, 032	$m_T = M_{\text{base}}/\xi$: one structure, sector-dependent base mass; range $= \xi \cdot \lambda_C(M)$ **[B]**
R90	Docs. 180, 019, 201	Exponent in $L_0 = \xi^1\ell_P$ derived from $m_T \sim 1/\xi$, not by analogy **[B]**
R91	Docs. 019, 201	λ in $m_T = \lambda/\xi$ is a base mass, not the Higgs quartic — label misleading **[K]**
R92	Docs. 250, 190	Doc. 250: ξ/ℓ is the coupling g_T, not m_T; Schwarzschild check for L_0 is an identity **[B]**
R93	Docs. 325, 329	$S_{\text{BH}}(M_{\text{coll}}) = 2\pi$ nat $= n_{\text{thresh}}$(Matzke) in bits; residual entropy of the endpoint **[B]**
R94	Doc. 325	$I_{\text{sector}}/I_{\text{thermal}} = \log_2 3$ for all M; structural constant of the $\mathbb{Z}_3$ geometry **[B]**
R95	Docs. 330, 270, 314, 327	Two wording refinements (flatness: $K{=}0$ pointwise via metric descent; s.a.: bounded *plus* symmetric); Type-I theorem [sketch]→**[B]** (function pullback p^*); D_4 specificity test open **[S]**
R96	Docs. 330, 332, 270	Type-I follow-up refinement after second Krüger feedback: $\text{Per}_{R_m}(\mathbb{R}_t)$ with per-period norm is isometrically isomorphic to $L^2(S^1_m)$, *not* to $L^2(\mathbb{R}_t)$ – pullback remains a change of representation, not a new sector **[B]**; Nash–Kuiper attribution in Doc. 330/R95(i) corrected (existence theorem for C^1, no C^2 non-embeddability claim) **[K]**
R97	Dok. 324	Vacuum operator correction following Doug Matzke's objection (IPI-Mail 26. Aug. 2026): Matzkes $V = N_1^+N_2^+N_3^+N_1^-N_2^-N_3^- = -64 \cdot P_{N1}P_{N2}P_{N3}$ is an involution in $\mathbb{Z}_3\mathbb{C}$ ($V^2 \equiv -V \pmod 3$) **[B]**; Dok. 324 incorrectly identified the Pp-product $P_{P1}P_{P2}P_{P3}$ (complementary sector, different ray: $\lvert\langle \text{vac}_V \vert \text{vac}_{Pp}\rangle\rvert = 0$) with Matzke's V; numerically verified by `pruef_324_vakuum_involution.py` (22 Assertions); Trine-Theorem $T_k^3 = \mathbf{1}$ und ξ-Nicht-Spektralwert remain correct **[B]**

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R98	Doc. 333, 332, 315, 295, 285	Two meanings of "unrolling" in the corpus distinguished and the recursion bound to the $\tilde{T} \cdot m$ duality (Doc. 333, 27 Aug. 2026): (i) Type-I (measurement act, lossy, prior corpus usage Doc. 285): $S^1_m \to \mathbb{R}_t$, winding number discarded; (ii) Type-III (canonical function pullback, lossless, Doc. 332/R96): $L^2(S^1_m) \cong \mathrm{Per}_{R_m}(\mathbb{R}_t)$, exact change of representation, m and t unroll together **[B]**. K_{frak} plays no role in the Type-III pullback; it sits in the choice of R_m via the mass formula and becomes visible only at the SI transition **[B]**. The recursion $\xi_{n+1} = \xi_n(1 - 100\,\xi_n)$ (Doc. 295/146) follows necessarily from $\tilde{T} \cdot m = 1$: every mass change forces a time change, which feeds back recursively. Case B (stable particle, $K_{\text{frak}} = 74/75$) is the projection of Case C (running recursion, log-spiral) onto the static approximation **[B]**. Doc. 332 reduced to a short cross-reference.
R99	Doc. 334, 307, 306, 285, 174	Superposition without time (Doc. 334, 27 Aug. 2026): (i) classical physics and QM inherit tacit instantaneity — time runs externally, calculations have no duration **[K]**; (ii) hidden inconsistency: state $\psi(t)$ is instantaneous, but probabilistic evaluation $\lvert\psi(t)\rvert^2$ requires real time for verification — this tension is absent from the formalism **[K]**; (iii) superposition is doubly timeless: no intrinsic "when", no verification without time expenditure; (iv) collapse has no duration in the QM formalism; (v) Pauli theorem does not apply to compact $\tilde{T}$ (angle variable); (vi) $\tilde{T} \cdot m = 1$ gives every mode an intrinsic time scale — superposition within one mode has scale $\tilde{T}_i$, cross-mode gives incompatible $\tilde{T}_i \neq \tilde{T}_j$ **[B]**
R100	Doc. 336, 324, 329	Notation updated to symmetric $\mathbb{Z}_3\mathbb{C}$ (after Matzke, IPI mail 28 Aug 2026): elements $a + bi$ with $a, b \in \{-1, 0, +1\}$ instead of $\{0, 1, 2\}$; $+1 + 1 = -1$ (ternary inline carry) **[E]**. Frobenius fixed points = $\{0, +1, -1\}$ = FFGFT sectors Δw **[B]**; norm classes $\{+1, -1\}$ instead of $\{1, 2\}$; tripotent eigenvalues $\{0, +1, -1\}$ **[B]**. Verification scripts 324–327 updated to symmetric $\mathbb{Z}_3\mathbb{C}$ (all assertions passed).
R101	Doc. 317, 336, 338	$\xi = 4/30000$ derived as Galois number **[K]**: core identity $(r_\mu/r_e)^2/\xi = \lvert\mathrm{GF}(9)^*\rvert^2 \cdot 5^2 \cdot \lvert\mathrm{GF}(27)\rvert = 43200$ (pruef_331). All six winding numbers $(n_{\theta,g}, n_{\phi,g})$ from GF(3^n) orders; recursion $n_{\theta,g} = n_{\theta,g-1} + n_{\phi,g-1}$ **[K]**.
R102	Doc. 338	$1/\alpha = 3700/27 = 137.037$ (dev. 7.6 ppm) from Galois group orders **[K]** (pruef_332): new observation $m_e \cdot m_\mu = 54 = \lvert\mathrm{GF}(3)^*\rvert \cdot \lvert\mathrm{GF}(27)\rvert$ MeV^2 (0.016 %; Doc. 011 gives geometric derivation). $\lvert\mathrm{GF}(27)\rvert$ cancels; no ξ, no v, no m_e as explicit input.

No.	Affects	Subject (short form; details: Archive)
R105	Doc. 006, 011, 070, 182, 231, 257, 285, 306, 307	**Retrospective marker certification — documents created before the marker system (1 Sept 2026).** These documents predate the introduction of the marker system (A010) and carry no markers of their own. Their derivations are internally consistent and are necessary chain links for the Galois programme (Docs. 336, 338, 339) and further documents. Retrospective status: Doc. 006 (particle masses, winding numbers r_i, p_i) **[K]**; Doc. 011 (fine structure α, E_0, fundamental relations) **[K]**; Doc. 231 (Hilbert space extension, L^2 structure) **[B]**; Doc. 257 (information unit, bit-energy scale) **[K]**; Doc. 285 (FFGFT-HLV dimension bridge) **[K]**; Doc. 306 (native time-energy reciprocity from $\tilde{T} \cdot m = 1$) **[K]**; Doc. 307 (time in state space) **[K]**; Doc. 182 (maximum universe scale from ξ, Schwarzschild radius and Hubble scale) **[K]**; Doc. 070 (mathematical structure: K_{frak} cancels exactly in ratios, absolute values carry the correction) **[B]**. Dependent documents may treat these results as a certified chain link by citing R105, without modifying the original documents.
R106	Docs. 342, 343	Primes forced by GF(3^k) **[B]**: the physical primes $\{5, 7, 11, 13\}$ are completely forced by the prime factorisations of $3^k - 1$ ($k = 3, 4, 5, 6$); $p = 13$ is the worldwide unique prime with $\text{ord}_p(3) = 3$. Euler factor threshold $p^* \approx 9.76$ $(s = 2) =$ 7-limit; Galois product density at chance level from $p_{\text{max}} \approx 30$ (pruef_342, pruef_344).
R107	Docs. 343, 338, 162, 186, 328	ξ as resolution floor **[B/S]**: the limit $k \approx 3$–4 (Theorem F) holds for $\varepsilon = 1.05\,\%$, one fractal level ($\approx 100\xi$, Doc. 162). At $\varepsilon = \xi$ (torus-mode linewidth, Doc. 328) the Galois grid stays resolved to $k \geq 8$, $p^* \approx 87$. Coupling: Farey $Q_c = \pi/\sqrt{6\varepsilon} \approx 12.5$; Arnold K^q vs. Farey $1/q^2$ — same tripartition, different law **[B]**; FFGFT with $K_{\text{eff}} = 2\pi\xi$ dynamically deeply subcritical, discreteness topological **[B]**. TFLN $Q > 10^6$ resolves ξ, not ξ^2. Doc. 186 preliminary note: locking **[B]**, Fibonacci **[B]**, B meson **[X]** corrected; renamed to `_De`/`_En`. Resonance factoriser capped at $N_{\text{max}} \approx 1/\varepsilon$ (13–26 bit); quantum computers/photonics as analogue computers with QM precision, curve at $N = 21$ since 2012 **[X]** (pruef_343c–e).
R108	Docs. 340, 342, 343	Dirac/Majorana partition of the GF$(27)^*$ classes **[B]**: sector pairing $k \mapsto -k$ and Majorana criterion $k \mapsto k^{-1}$ (mod 13) partition the four primitive classes identically — $\{f_1, f_2\}$ (orbits O_1, O_3, quadratic residues mod 13) Majorana, $\{f_3, f_4\}$ (O_2, O_4) Dirac. Canonical: Majorana layer = kernel of the quadratic character mod 13. Numbering unified per the polynomial table of Doc. 342 (sector pairing $f_1 \leftrightarrow f_2$, $f_3 \leftrightarrow f_4$; negation $f_i \leftrightarrow g_i$); exponent list in Doc. 342 corrected (f_3, f_4 were the same orbit). Consequence for Doc. 340: $\nu_1 \in O_1$, $\nu_2 \in O_3$ Majorana; $\nu_3 \in O_4$ **Dirac neutrino [B]** — no Majorana phase, m_{ee} from ν_1, ν_2 only (≤ 14.4 meV), inverted hierarchy more strongly excluded; sterile neutrinos drop out as a separate prediction. Doc. 340 revised directly (pruef_343f). Precursor: Doc. 007 derives $m_{\nu_1} = 4.54$ meV from $\xi^2/2 \times m_e$ **[K]** and motivates the double ξ suppression physically (neutrino = almost photon + geometric decoupling); flavour hierarchy first in Doc. 340. Addendum inserted in Doc. 007.

No.	Affects	Subject (short form; details: Archive)
R110	Docs. 320, 022	**Addenda** on ν_3 Dirac neutrino (Doc. 343 Theorem D‴, R108): Doc. 320 (spectral theory) — fixed-point derivation F_2, F_5, F_7 provides no Majorana/Dirac character; this follows from the Galois algebra of Doc. 340/343. Doc. 022 (QFT-ML addendum) — ξ^3 sterile suppression in DUNE row not to be interpreted as a sterile signal; Majorana layer fully occupied by ν_1, ν_2. Authoritative: Doc. 340/343.
R111	Doc. 344, 345, 346, 347, 348	**Charge quantisation, Gell-Mann–Nishijima, and generation structure from GF$(27)^*$ (4 Sept. 2026).** Doc. 344: Su[0]=Vss[0] exactly — neutrino = vacuum of Clifford-Fock space **[B]**; sector pairing $k \mapsto -k$ has no fixed point among primitive classes ⇒ all neutrinos Dirac **[B/S]**; nEXO (5–20 meV) decides. Doc. 345: literature comparison Furey/Dixon/Froggatt-Nielsen; FN: ε^{q_i} free vs. ξ^{p_i} forced **[K]**; after Docs. 346/347 equal to Furey/Dixon at charge content. Doc. 346: two Galois properties classify all SM fermions **[B]**: (i) quadratic residue mod 13 ⇔ electrically neutral; (ii) N mod 3 ⇔ colour charge; forbidden combination QR+triplet algebraically forced; Legendre symbol $(k/13)$ gives $Q \in \{0, \pm\frac{1}{3}, \pm\frac{2}{3}, \pm 1\}$ **[B]**. Doc. 347: $Q = I_3 + Y/2$ algebraically derived **[B]**: $Y = -1 + \frac{4}{3}$NQR from Legendre symbol (Doc. 346), $I_3 = \pm\frac{1}{2}$ from jE7 projector (Doc. 344); all four fermion charges exact. Theorem E **[S]** of Doc. 346 closed. Doc. 348: root orbits f_1–f_4 are genuine Frobenius orbits; 3 elements = 3 generations **[B]**. Trace bilinear form: $f_3 \times f_4$ (charged leptons × quarks) = permutation matrix (Gen.2 ↔Gen.3 forced) **[B]**; $f_1 \times f_3$: entries $1/\sqrt{2}$ **[B]**, consistent with large PMNS angles. CKM complexity: $\delta_{CP} \approx 68^\circ$ not derivable from GF$(27)^*$ ($\mathbb{Z}_3$ phases give 120°); usable as phase anchor (same status as $v = 246$ GeV). Check scripts: `pruef_346`, `pruef_347`, `pruef_348`, all assertions passed.
R109	Doc. 006 (restriction of R105)	**Neutrino part of Doc. 006 not tenable [X].** The "direct method" $E_i = 1/\xi_i$ is not a computation: $1/(\xi r_i)$ gives $e:\mu = 1:0.42$ (inverse hierarchy, dimensionless); the MeV values shown alongside are measured values. The neutrino masses $9.1/1.9/18.0$ meV follow from three incompatible prescriptions ($\nu_e = m_e\xi^2$; $\nu_\mu = r_\mu\xi^3 v$; ν_τ from none); the exponent $p_{\nu_\tau} = 25/9$ is r_τ copied into the exponent column. The Yukawa method $m_i = r_i\xi^{p_i}v$ remains valid for charged leptons and quarks ($< 1.2\,\%$). The **[K]** certification in R105 applies to Doc. 006 only for r_i, p_i and the Yukawa method, not for neutrinos and not for the claimed equivalence of both methods. Authoritative for neutrinos: Doc. 340 (R103, R108). Doc. 343 D‴ records the discrepancy **[S]**.
R112	Doc. 006, 319, R104; Doc. 351	**$v = 246$ GeV is not a measurement but an SM definition [Q] (5 Sept. 2026).** $v \equiv (\sqrt{2}\,G_F)^{-1/2}$ follows from $M_W = gv/2$ and $G_F/\sqrt{2} = g^2/(8M_W^2)$; only $\tau_\mu = 2.196\,980\,3(22)$ µs (MuLan) is measured, G_F follows via $\tau_\mu^{-1} = G_F^2 m_\mu^5/(192\pi^3)\,(1+\Delta q)$ with $\Delta q = -0.44\,\%$ (QED $O(\alpha^3)$, hadr. VP with 4.7σ CMD-3 tension; Eberhart et al. 2026) and Sirlin absorption of EW corrections. The irrational factors $192\pi^3$ and $\sqrt{2}$ come from the SM formulation, not from measurement. Corpus-wide wording correction: "$v = 246$ GeV (measured)" → "$v = 246$ GeV (SM definition from G_F)"; the FFGFT value $v = 245.6$ GeV (R104) is a comparison of structure against convention. Convention-free test only at the τ_μ level — **open [S]**.

No.	Affects	Subject (short form; details: Archive)
R113	Doc. 338, 317, 011, R100, R101; Doc. 351	**Anchor m_μ/m_e is QED-dependent; comparison values are not theory-free [Q] (5 Sept. 2026).** $m_\mu/m_e = 206.768\,277(24)$ derives solely from muonium HFS (LAMPF 1999, Liu et al.); extraction via $\nu_F = \frac{16}{3}\alpha^2 c R_\infty (m_e/m_\mu)(m_r/m_e)^3$ — the anchor implicitly contains α^2; Eides (2026): CODATA underestimates the theory uncertainty by a factor of 5–10 (271–515 Hz instead of 51 Hz). α: Rb (Morel 2020) and Cs (Parker 2018) disagree at $> 5\sigma$; the CODATA average covers neither. m_τ: two methods (threshold scan BESIII 1776.91, pseudomass Belle II 1777.09 with $m_{\nu_\tau} = 0$) differ by 0.18 MeV; PDG average $1776.93(9)$, FFGFT 1776.97 lies 0.4σ — test undecided. Exact computation: the residual $43200 - 42753.12 = 446.9$ in $(m_\mu/m_e)^2$ is $240\,000\sigma$ of the CODATA uncertainty; no measurement error explains it; it is an empirical discrepancy **[K]** (Doc. 352, § 9–10); physical attribution (R113 bridge) open **[S]**. Wording correction: "model-independently measured" → "least model-dependent available value". Bookkeeping rule: every prediction states anchor, number of steps, corrections applied, extraction method of the comparison value. All 21 sources verified (Doc. 351) Three Galois identities compared in rational arithmetic (verification script `pruef_351_rationale_vergleiche.py`): (1) $(m_\mu/m_e)^2 = 43200$: residual $-1.034\,\%$ ($240\,000\,\sigma$), physical **[B]**. (2) m_μ/m_e vs. $\sqrt{43200} = 120\sqrt{3}$: residual $-0.52\,\%$ is *not an independent finding* — $\sqrt{43200}$ is irrational; the residual is exactly half the quadratic residual (artefact of taking the square root). (3) $m_e{\cdot}m_\mu = 54$ MeV^2 ($= \lvert GF(3)^*\rvert \cdot \lvert GF(27)\rvert = 2 \cdot 27$): residual $-0.016\,\%$ ($7386\,\sigma$), physical, 65 times smaller than the quadratic residual; no computational artefact (rounding uncertainty 2.2% of measurement uncertainty), not covered by measurement tolerance **[K]**. **Bookkeeping rule:** Galois identities should be compared only in their natural algebraic form — where no irrational numbers appear.

No.	Affects	Subject (short form; details: Archive)
R114	Doc. 351 (update), Doc. 352 (new); R113	**Lepton residuals: lepton ladder, Galois and Koide compared; R113 attribution clarified [Q] (5 September 2026).** Doc. 352 compares three FFGFT structures (lepton ladder Docs. 006/317, Galois identities Doc. 351, Koide relation) with PDG and expresses all residuals in units of ξ. Main findings: (1) Koide matches PDG to $0.046 \cdot \xi$ — most precise prediction **[K]**. (2) Galois difference residual $-77.6 \cdot \xi = -2 \times (+39.1 \cdot \xi)$ lepton ladder algebraically forced **[B]**. (3) Generation-dependent corrections ε_i (order 73–$192 \cdot \xi$, decreasing with generation) explain transition $Q_{\text{bare}}(+8 \cdot \xi) \to Q_{\text{PDG}}(-0.05 \cdot \xi)$ to 101 % **[K]**. (4) Z_3 circulant: angle $\theta = 2/9$ from Doc. A110 **[K]**; amplitude $d/c = \sqrt{2}$ (Koide condition) and θ are orthogonal parameters **[B]** — $Q = \frac{1}{3} + \frac{d^2}{6c^2}$ does not contain θ. (5) Status separation of the ε_i (clarified 6 September 2026): $\varepsilon_i = m_i^{\text{PDG}}/m_i^{\text{bare}} - 1$ is a *computed residual* **[K]** from PDG **[Q]** against FFGFT bare **[K]** — not a number taken from QED literature; its ξ scaling is observed **[K]**; the assignment to QED extraction + SI projection is an *attribution* **[S]**, not a citation; the reason for the falling structure $\varepsilon_e > \varepsilon_\mu > \varepsilon_\tau$ is open **[S]**. The QED self-energy formula $\delta m/m \approx \frac{3\alpha}{4\pi} \ln(m/\mu)$ in Doc. 352 § 10 is **[Q]** (standard formula) but yields only the order of magnitude, not the ε_i values themselves. Column “Origin” in Doc. 352 § 9 → “Presumed origin [S]”. Doc. 351 updated: three occurrences of “physical (R113)” → “empirical discrepancy [K] (Doc. 352, § 9–10); physical attribution (R113 bridge) open [S]”. Verification script `pruef_352_leptonreste.py` (32/32) **[K]**.

.2 Currently open bridges (as of R113)

- τ_μ directly from FFGFT, without detour via G_F and v — the only convention-free test of the weak sector (R112, Doc. 351) [S]
- Why do ε_i decrease with generation? The ε_i are computed residuals **[K]**; their attribution to QED extraction + SI projection is **[S]**; whether the QED self-energy is derivable from the T^4/Z_3 geometry is open — R113 bridge (Doc. 352, § 10) [S]
- Koide amplitude $d/c = \sqrt{2}$ from FFGFT geometry — orthogonal to $\theta = 2/9$ **[B]**, not yet anchored (Doc. 352, § 8) [S]
- Charge quantisation $Q \in \{0, \pm\frac{1}{3}, \pm\frac{2}{3}, \pm 1\}$: *closed by Doc. 346, R111 [B] —*
- Gell-Mann–Nishijima $Q = I_3 + Y/2$: *closed by Doc. 347, R111 [B] —*
- Δm^2_{32} atmospheric ($+16.0\,\%$): *closed by Doc. 340, R103 [K] —*
- 2-loop corrections to $\alpha_s(M_Z)$ [S]
- m_{Pl} and $\alpha_{\text{em}}(M_Z)$ from ξ (external inputs in R81) [S]
- Quark/hadron sector — Doc. 318, R76
- Greybody factors; back-reaction on orbifold fixed points — R85 [S]
- Forward derivation of the cosmic exponent $41/4$ (P20); the 10^{123} question, $R_H \to \ell_1$ and $\kappa = \Lambda_{\text{obs}} R_H^2$ hang on it
- CMB peak selection: generate $\{1, 6, 14, 26\}$ forward; the $|n|^2{=}30$ case (P29/P31)
- C_ℓ projection: physical source/window function (P30)

- Ω_{DM} quantitatively, model-neutral (P17)
- Spectral dimension $d_s = 1.86$ vs. FFGFT ≈ 2
- Corpus-wide H_0 language cleanup (P34); fundamental rework of Doc. 026 (P16)
- Justify the base mass $M_{\text{base}} = m_P$ in the fundamental sector — Doc. 180, R90 [S]
- ξ as line width (P44, pre-registered); ι embedding HLV$\subset$FFGFT in the time sector (Doc. 297)
- Adversarial D_4 operator-specificity test against matched comparison carriers (GAE logic) — R95 [S]

R86 – Self-adjointness of $\hat{F}$ (August 2026): Doc. 327 closes the gap declared as [S] in R82 completely. The finiteness of the scale ladder is guaranteed by $L_0 = \xi \cdot \ell_P$ (Doc. 180 **[K]**) – no infinities in FFGFT, hence $\hat{F}$ is bounded and $\mathscr{D}(\hat{F}) = \mathscr{H}$. The symmetry $\hat{F} = \hat{F}^\dagger$ follows from the $\mathbb{Z}_3$ pairing $(k, -k)$ (the same as in Doc. 325 **[B]**). Deficiency indices $(0, 0)$: exactly one self-adjoint realisation **[B]**. Fractal measure (finite 100-step recursion), $\mathbb{Z}_3$ restriction and all three χ-twist classes preserve self-adjointness **[B]**. $\xi = \lambda_{\min}(\hat{F}_{D_4})$ well-defined and truncation-stable (min-max) **[B]**. No remaining cases. Verification script: `327_selbstadjungiertheit_F_verify.py` (21 assertions).
Declared 23 August 2026

R87 – Δm^2_{32}: corrected numerical values and candidate analysis (August 2026): In Doc. 320 the evaluation of $\xi^{9/5}$ was faulty (8.717×10^{-8} instead of 1.059×10^{-7}); the mass formula $m_{\nu_i} = m_e\,\xi^{p_i}$ and the exponents $p_i \in \{9/4,\ 2,\ 9/5\}$ were always correct. Docs. 320 and 322 have been brought to the correct numerical values: $m_{\nu_3} = 54.11$ meV **[K]**, $\Delta m^2_{32} = 2.846 \times 10^{-3}$ eV² **[K]** (deviation $+16.0\,\%$ against NuFIT 5.3), $\sum m_\nu = 64.17$ meV **[K]** (still within the Planck limit). Unaffected: m_{ν_1}, m_{ν_2}, Δm^2_{21} $(+8.3\,\%)$ and the fixed-point assignment. Doc. 320 was at the same time extended by a candidate analysis for the remaining deviation: K_{frak} at fixed point F_7 ($+12.9\,\%$, insufficient), mixing term F_5–F_7 (right sign, complete mass matrix outstanding – the most promising candidate), alternative exponents (unlikely: $p_3^{\text{PDG}} = 1.8081$ lies only $0.45\,\%$ above $9/5$). Δm^2_{32} remains **[S]**. Verification script: `320_dm32_neutrino_verify.py` (10 assertions).
Declared 23 August 2026

R88 – n_{thresh} is scale-dependent; Matzke's scale reconstructed (August 2026): Doc. 329 settles numerically the question booked as [S] in Doc. 326. The collapse condition Compton = Schwarzschild yields $M_{\text{coll}} = m_P/\sqrt{2}$ purely geometrically and free of any bit value **[B]**. With the system-dependent bit energy $E_{\text{bit}}(L) = \hbar c/L$ (Doc. 257) one obtains exactly $n_{\text{thresh}}(L) = L/(\sqrt{2}\,\ell_P)$ – linear in the scale **[B]**. Matzke's value 6.4097 corresponds exactly to $L = 2\pi\ell_P/\ln 2 = 9.0647\,\ell_P$ **[B]**; the "universal" bit value is the FFGFT bit energy at that scale. Across the scales of the corpus n_{thresh} varies by a factor of 4.6×10^{29}; a $10\,\%$ temperature deviation flips the stability claim for Cl(6). Hence $n_{\text{thresh}} = 6.41$ as a universal number is negatively closed **[X]**. Untouched: Compton = Schwarzschild, r_s without field equations, Cl(6) structure **[B]**. Verification script: `326_nthresh_skalenabhaengigkeit.py` (12 assertions).
Declared 24 August 2026

R89 – Time-field mediator mass: one structure, sector-dependent base mass (August 2026): The corpus uses m_T with widely differing numerical values (Docs. 019/180/201: $m_T = \lambda/\xi$; Doc. 032: effective torsion mass ≈ 5.220 GeV). This is no contradiction: both follow the rule $m_T = M_{\text{base}}/\xi$, whence the range of the massive time field is $\hbar/(m_T c) = \xi \cdot \lambda_C(M_{\text{base}})$ **[B]** — universal form, sector-dependent value, structurally identical with $E_{\text{bit}} = \hbar c/L$ (Docs. 257, 329). Lepton sector ($M = m_e$): $m_e/\xi = 3.83$ GeV, with the $O(1)$ factors of Doc. 032 ($\sin\pi\xi$, π^2, $\sqrt{\alpha/K_{\text{frak}}}$, R_f) 5.220 GeV — range 3.78×10^{-17} m. Fundamental sector ($M = m_P$): $m_T = m_P/\xi = 7500\,m_P$ — range 2.155×10^{-39} m $= L_0$ exactly. Doc. 032 itself calls its quantity an

effective torsion mass; sameness of name is not sameness of quantity. Verification script: `180_mT_struktur_L0_verify.py` (17 assertions).
Declared 24 August 2026

Placement after Doc. 287: The Lagrangian description ($m_T = M_{\text{base}}/\xi$) and the bit-energy description ($E_{\text{bit}} = \hbar c/L$) are *different representations of the same floor structure* – in the terms of Doc. 287, presumably a **bijection**: lossless and mutually translatable, but neither more complete than the other. The generative core that loses nothing is the T^4 torus (Doc. 287, Doc. 243); all other descriptions stand to it in one of three relations: identity, bijection or projection.

R90 – The exponent in $L_0 = \xi \cdot \ell_P$ is derived (August 2026): Doc. 180 justifies $L_0 = \xi \cdot \ell_P$ by the ξ scaling of the coupling term $g_T^\ell = \xi m_\ell$ — an argument by analogy. The exponent however follows *necessarily* from the Lagrangian mass term: $m_T = M_{\text{base}}/\xi^1$ together with $\lambda_C \sim 1/m$ gives range $\sim \xi^1$ **[B]** (R89). Counter-check: $m_T \sim \xi^{-n}$ would give $L_0 = \xi^n \ell_P$; the exponent of the range is the negative exponent of m_T. L_0 is thus the Compton range of the time-field mediator in the fundamental sector — below it the time field can no longer mediate, which is why "distance" loses its meaning. This also explains *why* a floor exists at all: because the mediator is massive. What remains open is only the choice $M_{\text{base}} = m_P$ for the fundamental sector; it introduces no new parameter ($\lambda = 1$ in Planck units) but is not independently justified [S]. Without consequences for Doc. 327 (R86): boundedness requires only a finite floor **[B]**.
Declared 24 August 2026

R91 – λ in $m_T = \lambda/\xi$ is a base mass (August 2026): Docs. 019 and 201 describe λ as the "Higgs coupling parameter". This is misleading: (i) Dimensionally λ must be a *mass*, since m_T is a mass and ξ is dimensionless **[B]**; the Higgs quartic $\lambda_h \approx 0.129$ is dimensionless. (ii) λ_h runs under the RGE and is a low-energy parameter of the electroweak scale; a value measured there cannot be inserted at the Planck scale. (iii) Interpreted as a dimensionless factor in Planck units, λ_h would give the range $7.75 \cdot L_0$ — missing L_0. Correct is $\lambda = M_{\text{base}}$ of the respective sector, in the fundamental case m_P (R89/R90). A Higgs connection can at most be an effective statement in the electroweak sector, never the definition. The label in Docs. 019/201 is to be corrected [S].
Declared 24 August 2026

R92 – Two misattributions concerning the time field (August 2026): (i) *Doc. 250*: The sentence "The term $-\frac{1}{2}m_T^2(\partial m)^2$ with $m_T = \xi/\ell$ (Doc. 201)" contains two errors. In natural units $m = 1/\ell$, so $\xi/\ell = \xi \cdot m$ — this is the *coupling* $g_T^\ell = \xi m_\ell$, not the mediator mass $m_T = M_{\text{base}}/\xi$ **[B]**. Furthermore the mass term reads $-\frac{1}{2}m_T^2(\Delta m)^2$, not $(\partial m)^2$; the latter would contribute to the kinetic term. The physical point (the mass term provides the stiffness against divergence of Δm) remains correct; only the formula is misquoted [S]. (ii) *Doc. 190, Clarification 4*: The statement recorded as an internal consistency check, that an object with $M_0 = \xi m_P/2$ would have Schwarzschild radius exactly L_0, is an *identity*: from $r_s = 2GM/c^2$ and $\ell_P = Gm_P/c^2$ it follows that $r_s(x m_P/2) = x\ell_P$ for any x and any exponent **[B]**. It tests neither ξ nor the exponent in L_0; classifying it as a "check" overstates it. Non-trivial would be only the condition appended there, whether M_0 coincides with a rung of the ξ hierarchy — open [S].
Declared 24 August 2026

R93 – $S_{\text{BH}}(M_{\text{coll}}) = 2\pi$ nat $= n_{\text{thresh}}$ (Matzke) in bits (August 2026): Doc. 325 §5.3 (new) and Doc. 329 **[B]** jointly yield: $r_s(M_{\text{coll}}) = \sqrt{2}\,\ell_P$ (geometric, Compton = Schwarzschild); $S_{\text{BH}}(M_{\text{coll}}) = 4\pi \cdot 2\ell_P^2/(4\ell_P^2) = 2\pi$ nat exactly. In bits: $2\pi/\ln 2 = 9.0647$ bit $= n_{\text{thresh}}$ (Matzke) **[B]**. Interpretation: Matzke's stability threshold is the residual entropy of the evaporation endpoint — the same geometry in two languages. Matzke's universality claim remains closed negative **[X]** (R88); this identity explains *why* the value 9.065 appears in both contexts

without making it universal. Verification script: `325_informationsbilanz.py` (9 assertions, assertions 2–3).
Declared 25 August 2026

R94 – $I_{\text{sector}}/I_{\text{thermal}} = \log_2 3$ for all M (August 2026): From Doc. 325 §5.1 **[B]** ($\log_2 3$ bits of sector information per quantum, orthogonal $\mathbb{Z}_3$ pair states) and §Building Block 2 **[B]** (1 nat thermal area bookkeeping per quantum via Clausius):

$$\frac{I_{\text{sector}}}{I_{\text{thermal}}} = \log_2 3 \approx 1.585 \quad \text{for every mass } M > M_{\text{coll}}.$$

The ratio is mass-independent — a structural constant of the $T^4/\mathbb{Z}_3$ geometry. Total budget: $N_{\text{emit}} = S_{\text{BH}}(M_{\text{init}}) - 2\pi\,\text{nat}$, $I_{\text{sector}} = N_{\text{emit}} \cdot \log_2 3\,\text{bits}$. Verification script: `325_informationsbilanz.py` (assertions 6–7).
Declared 25 August 2026

K-041-a/b/c – Internal inconsistencies in Doc. 041 (August 2026): Doc. 041 remains unchanged. The following notes document places where printed formulas and table values in the same document disagree under the document's own $\xi = (4/3) \cdot 10^{-4}$ (Doc. 328 §7.4b). Resolution requires author review of the source document.

K-041-a (δ_{CKM}): Formula $\delta_{\text{CKM}} = \arcsin\big(2\sqrt{2} \cdot \sqrt{\xi}/3\big)$ yields 0.0109 rad; table value 1.20 rad. Candidate: $\arcsin\big(2\sqrt{2}/3\big) = 1.231$ rad — factor $\sqrt{\xi}$ likely a typesetting error.

K-041-b (θ_{13}): Formula $\theta_{13} = \arcsin\big(\xi^{1/3}\big)$ yields $2.93°$; table value $8.57°$. Candidate: argument $1/300 = 25\xi$ yields $8.59°$ — whether $1/300$ is a corpus quantity is to be clarified at the source document.

K-041-c ($|V_{us}|$): Formula with Doc. 006 masses yields 0.2124; table value 0.22452 (dev. $5.4\,\%$). Possible cause: differing masses or f_{Cab} definition in Doc. 041.
Declared 25 August 2026

R95 – Doc. 330: two wording refinements and Type-I clarification after external review (August 2026): External review (M. Krüger, IPI, 25 August 2026) of Doc. 330 identified two overly superficial formulations; the corpus itself (Docs. 270/314/327, R41, R86) was correct and more precise in each case. Doc. 330 remains unchanged; the refinements apply at the register level.

(i) Intrinsic flatness of T^4 (Doc. 330 §2.2). **Too superficial:** "positive and negative curvature contributions cancel in the sum" — this conflates the *embedded* torus in $\mathbb{R}^3$ ($K > 0$ outside, $K < 0$ inside, only globally $\int K\,dA = 0$ via Gauss–Bonnet, $\chi = 0$) with the *flat* torus $\mathbb{R}^4/\mathbb{Z}^4$. **Precise:** The flat metric δ_{ij} is $\mathbb{Z}^4$-invariant and descends undistorted to the quotient; $\Gamma^k_{ij} = 0$, Riemann tensor pointwise zero, $K = 0$ everywhere — there is nothing to cancel. Triply verified constructively ($\Gamma = 0$, angle sum π, trivial holonomy). The conclusion of Doc. 330 (intrinsically flat, Fourier basis without curvature corrections, Type III bijective) remains unchanged **[B]**. Nash–Kuiper side note: the flat torus is not C^2-isometrically embeddable in $\mathbb{R}^3$ — the embedding question does not arise for the fundamental level anyway.

(ii) Self-adjointness of $\hat{F}$ (Doc. 330 §Question 3). **Too superficial:** "bounded and therefore self-adjoint" — boundedness alone does not imply self-adjointness (counterexample: shift operator). **Precise (= R86 [B]):** bounded *plus* symmetric via the $\mathbb{Z}_3$ pairing $(k, -k)$ on a complete Fourier basis ⇒ self-adjoint; deficiency indices $(0,0)$; boundedness ($\|\hat{F}\| \leq \sum_n \xi^n = \xi/(1-\xi)$, convergent even for $N \to \infty$) secures $\mathcal{D}(\hat{F}) = \mathcal{H}$, while the symmetry argument carries the burden independently. Status **[B]** unchanged.

(iii) Type-I direction clarification (no error; confirmation of R41). The review objection that the unrolling $S^1_m \to \mathbb{R}_t$ has a lifting problem (covering $p : \mathbb{R} \to S^1$ without canonical inverse) concerns the *lifting of points* — Type I, however, lifts no points but is the *function pullback* $\Phi = p^* : L^2(S^1_m) \to \text{Per}_{R_m}(\mathbb{R}_t)$, $(\Phi f)(t) = f(t \bmod R_m)$ — canonical, no additional datum. Precisely the reverse direction $\mathbb{R} \to S^1$ is the lossy one (R41, declared 19 June

2026). The external six-point checklist is answered completely: injective; surjective onto Per_{R_m} (not onto $L^2(\mathbb{R}_t)$ — that *is* the embedding statement of Doc. 270); isometry per period (Haar ↔ Lebesgue/period); inverse = restriction to an arbitrary period (period choice = deck-transformation gauge freedom, not a missing datum); spectrum discrete ↔ discrete frequencies n/R_m; $\mathrm{U}(1)$ translation ↔ time translation commute with Φ. The proof sketch of Doc. 270 (theorem "Type I is information-preserving in mode content") is thereby completed to a full theorem: status change [sketch] → **[B]**.

(iv) Accepted as open: An adversarial operator-specificity test for D_4 against matched comparison carriers (GAE logic) does not exist; $5 \nmid |\mathrm{Aut}(D_4)| = 1152$ is a lattice-specific algebraic fact (excluding e.g. E_8 with $5 \mid |\mathrm{Aut}(E_8)|$), not a specificity proof. Carried as an open bridge **[S]**.

Verification scripts: `pruef_krueger_einwaende.py` (16/16), `pruef_k1_tief.py` (8/8), `pruef_k3_tief.py` (6/6), `pruef_k4_tief.py` (5/5).
Declared 25 August 2026

R96 – Type-I follow-up refinement: pullback is a change of representation, not a new sector (August 2026): Second feedback (M. Krüger, IPI, 26 August 2026) on the R95(iii) clarification accepts the function pullback $\Phi = p^*$ as the correct mathematical object and the refutation of the original lifting objection, but refines a remaining point: Φf is R_m-periodic, so $\int_{\mathbb{R}} |\Phi f|^2\, dt = \infty$ for any $f \neq 0$ — Φf does not lie in ordinary $L^2(\mathbb{R}_t)$ with Lebesgue measure over the whole line. The correct target space $\mathrm{Per}_{R_m}(\mathbb{R}_t)$ (periodic L^2_{loc}, norm over one period) is isometrically isomorphic to $L^2(S^1_m)$ itself — the same countable Fourier basis $\{e_n\}_{n\in\mathbb{Z}}$, the same Parseval identity. Formally:

$$L^2(S^1_m) \;\cong\; \mathrm{Per}_{R_m}(\mathbb{R}_t) \quad \text{(per-period norm)}, \qquad \textit{not} \qquad L^2(S^1_m) \;\cong\; L^2(\mathbb{R}_t).$$

Algebraically confirmed (`pruef_332_krueger_periodizitaet.py`): divergence of the whole-line integral for any nontrivial periodic function (translation-invariant Lebesgue measure, each period contributes the same positive amount); isomorphism of the Fourier bases of both spaces via $\langle e_{n_1}, e_{n_2}\rangle = R_m\, \delta_{n_1 n_2}$ over one period. Status **[B]**.

Assessment: This is not a correction to Doc. 330/R95(iii), but a precisification of what the status change [sketch]→[B] there mathematically carries: an information-preserving representation of the compact sector on the universal cover, not a genuinely noncompact state space. Doc. 332 (Section 2, "canonical function pullback") already formulated this restraint correctly ("change of representation of the same function algebra", no extension), but without explicitly separating the norm/Hilbert-space level from the algebra level; Doc. 332 was accordingly extended with a clarifying paragraph (Section 2.1, "continuum of the coordinate representation" vs. "continuum of the state space") and recompiled.

Side correction (Nash–Kuiper): The Nash–Kuiper side note in Doc. 330/R95(i) is refined: Nash–Kuiper is an existence theorem for C^1-isometric embeddings, not a non-existence theorem for C^2 embeddings of the flat 2-torus in $\mathbb{R}^3$ — these two statements (existence of C^1 isometries and the C^2 obstruction question) must be kept separate. For the fundamental level T^4 the embedding question into $\mathbb{R}^3$ does not arise dimensionally anyway; the point was not load-bearing for the main R95(i) conclusion (intrinsic flatness) and does not change its status **[B]**.

Verification script: `pruef_332_krueger_periodizitaet.py` (4/4 assertions).
Declared 26 August 2026

Declared 25 August 2026